

## Lecture 12: Regression Inference

From Fitting Lines to Testing Coefficients

## Previously, on Lectures 10 & 11...

***t*-tests:** Compare one or two means. Welch's for unequal variances.

$\chi^2$ : Goodness-of-fit and independence for counts/categories.

**ANOVA:** Compare  $k$  group means via  $F = MS_B/MS_W$ .

**A/B testing:** Same hypothesis testing, different engineering (peeking, MDE).

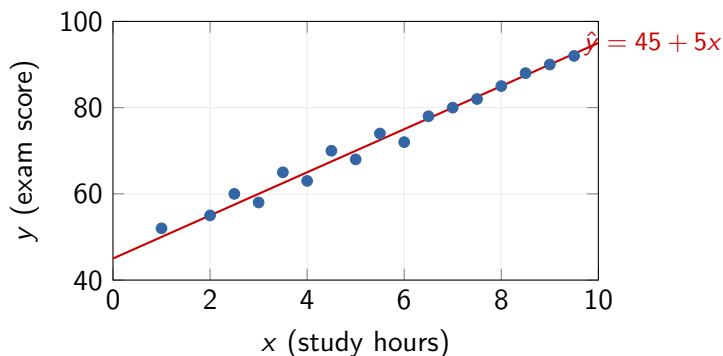
**All tests:** Assume some model, measure evidence against  $H_0$ .

**Today:** The most widely used model in statistics: **linear regression**.  
Not just fitting a line — testing *which predictors matter* and *how much*.

# Part I: The Linear Model

From the math module to the statistical framework

## Quick Recall: OLS Regression



**Math module:** Find  $\hat{\beta}_0, \hat{\beta}_1$  that minimize  $\sum (y_i - \hat{y}_i)^2$ .

**Statistics question:** Is  $\beta_1$  *really* nonzero, or could the slope be noise?

# The Statistical Model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i$$

where  $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

$\beta_0, \beta_1, \dots, \beta_p$

True (unknown)  
population parameters.

*Fixed* numbers.

$\varepsilon_i$

Random error.  
Everything the model  
doesn't capture.

$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$

OLS estimates.  
*Random* — they vary  
from sample to sample.

**Key shift:**  $\hat{\beta}$  has a *sampling distribution*, just like  $\bar{X}$  from L7.

# Assumptions of the Linear Model

**1. Linearity:**  $E[Y | X] = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ .

Check: residuals vs fitted plot should show no pattern.

**2. Independence:** Errors  $\varepsilon_i$  are independent of each other.

Violated by: time series, clustered data, spatial data.

**3. Homoscedasticity:**  $\text{Var}(\varepsilon_i) = \sigma^2$  (constant variance).

Check: residuals vs fitted — fan shape = heteroscedasticity.

**4. Normality:**  $\varepsilon_i \sim N(0, \sigma^2)$ .

Check: QQ plot of residuals. Least important for large  $n$  (CLT helps).

Mnemonic: **LINE** — Linearity, Independence, Normality, Equal variance.

# Part II: Inference for Coefficients

SE,  $t$ -tests,  $p$ -values, and CIs for each  $\beta_j$

# The Sampling Distribution of $\hat{\beta}$

OLS estimates are random variables with known distributions

Under the linear model assumptions:

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 (X^\top X)^{-1}_{jj}\right)$$

$\hat{\beta}_j$  is **unbiased**:  $E[\hat{\beta}_j] = \beta_j$ . Variance depends on  $\sigma^2$  and the design matrix  $X$ .



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$$\text{Standard error of } \hat{\beta}_j: \text{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{(X^\top X)^{-1}_{jj}}$$

$$\text{where } \hat{\sigma}^2 = \frac{\sum e_i^2}{n - p - 1} = \frac{\text{RSS}}{n - p - 1} \quad (\text{residual standard error})$$

This is just  $\text{SE} = (\text{estimated SD}) / (\text{effective } \sqrt{n})$ , same idea as L7.

# Simple Linear Regression: SE Formula

For  $Y = \beta_0 + \beta_1 X + \varepsilon$ , the formulas simplify

$$\begin{aligned} \text{SE}(\hat{\beta}_1) &= \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}} \\ \hat{\sigma} &= \sqrt{\text{RSS}/(n-2)} \end{aligned}$$

## SE is small when:

- Residuals are small ( $\hat{\sigma}$  small)
- $X$ 's are spread out
- Sample size  $n$  is large

## SE is large when:

- Lots of noise ( $\hat{\sigma}$  large)
- $X$ 's are clustered together
- Sample size  $n$  is small

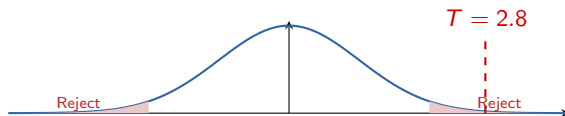
Intuition: a slope is hard to estimate when  $X$  doesn't vary much (imagine all points at  $X = 5$ ).

## t-Test for a Single Coefficient

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0$$

$$T = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim t_{n-p-1}$$

under  $H_0$



**Interpretation:** “Is this predictor useful, *controlling for* all others in the model?”  
 $H_0: \beta_j = 0$  means  $X_j$  has no linear effect on  $Y$  after accounting for other predictors.

## Example: Predicting House Prices

$n = 100$  houses. Model:  $\text{Price} = \beta_0 + \beta_1 \cdot \text{Size} + \beta_2 \cdot \text{Bedrooms} + \beta_3 \cdot \text{Age} + \varepsilon$ .

|              | Estimate | SE    | $t$   | $p$ -value |
|--------------|----------|-------|-------|------------|
| (Intercept)  | 50.2     | 15.3  | 3.28  | 0.001      |
| Size (sq ft) | 0.12     | 0.015 | 8.00  | $< 0.001$  |
| Bedrooms     | 5.1      | 4.8   | 1.06  | 0.291      |
| Age (years)  | -0.85    | 0.22  | -3.86 | $< 0.001$  |

**Size:**  $t = 8.0$ ,  $p < 0.001$   
Strong evidence. Each sq ft adds \$120 (significant).

**Bedrooms:**  $t = 1.06$ ,  $p = 0.29$   
Not significant. After controlling for size, bedrooms don't add much info.

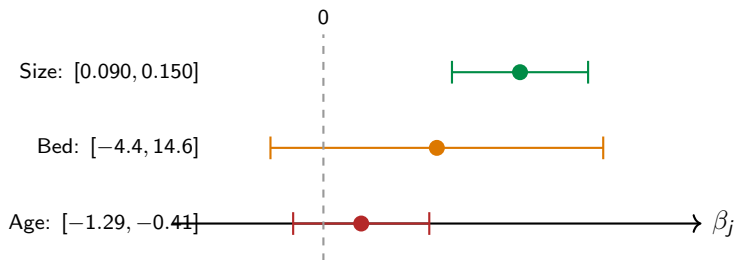
**Age:**  $t = -3.86$ ,  $p < 0.001$   
Significant. Each year reduces price by \$850.

## Confidence Intervals for Coefficients

$$\hat{\beta}_j \pm t_{n-p-1, \alpha/2} \cdot \text{SE}(\hat{\beta}_j)$$

Same Wald CI formula from L8, applied to regression coefficients.

**House price example** (95% CI,  $t_{96, 0.025} \approx 1.985$ ):



Bedrooms CI contains 0  $\Rightarrow$  not significant. Size and Age CIs exclude 0  $\Rightarrow$  significant.

**CI and  $p$ -value always agree:** CI contains 0  $\Leftrightarrow p > \alpha$ .

# Part III: Overall Model Fit

Is the model as a whole useful? How much does it explain?

# The Overall $F$ -Test

Tests whether *any* predictor is useful

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \quad (\text{no predictor matters})$$

$$F = \frac{(SS_T - \text{RSS})/p}{\text{RSS}/(n - p - 1)} = \frac{MS_{\text{Reg}}}{MS_{\text{Res}}} \sim F_{p, n-p-1}$$

**This is exactly ANOVA applied to regression!**

- $SS_{\text{Total}} = \text{total variation in } Y$
- $SS_{\text{Regression}} = SS_T - \text{RSS} = \text{variation explained by the model}$
- $\text{RSS} = \text{residual sum of squares} = \text{unexplained variation}$

Same logic: if the model explains much more than noise,  $F$  is large  $\Rightarrow$  reject  $H_0$ .

**House prices:**  $F = 85.3$  on 3 and 96 df,  $p < 0.001$ . The model is highly significant.

## $R^2$ and Adjusted $R^2$

$$R^2 = 1 - \frac{\text{RSS}}{\text{SS}_T}$$

Proportion of variance  
in  $Y$  explained by model.

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{SS}_T/(n - 1)}$$

Penalizes for adding  
useless predictors.

$R^2 < 0.3$   
**Weak**

$R^2 \approx 0.5$   
**Moderate**

$R^2 \approx 0.7$   
**Good**

$R^2 > 0.9$   
**Excellent**

**Trap:**  $R^2$  *always* increases when you add predictors, even useless ones.  
Use  $R^2_{\text{adj}}$  to compare models with different numbers of predictors.  
“I got  $R^2 = 0.95$ ” means nothing if you have  $n = 20$  and  $p = 18$ .



## Part IV: Why OLS?

The Gauss–Markov theorem: OLS is the best you can do (with conditions)

# The Gauss–Markov Theorem

## Gauss–Markov Theorem

Under the assumptions of linearity, independence, homoscedasticity, and  $E[\varepsilon] = 0$   
(normality **not** required!),

OLS is **BLUE**: Best Linear Unbiased Estimator.

Among all *linear, unbiased* estimators of  $\beta$ , OLS has the *smallest variance*.

**Best**  
smallest variance

**Linear**  
 $\hat{\beta} = AY$  for some  
matrix  $A$

**Unbiased**  
 $E[\hat{\beta}] = \beta$

**Estimator**  
it's a function  
of the data

*Connects to L3:* among unbiased estimators, lowest variance = best. Same principle as Cramér–Rao.

## When Gauss–Markov Doesn't Help

**Heteroscedasticity:** Variance of  $\varepsilon$  changes with  $X$ .

**Fix:** Use **robust (HC) standard errors** — same  $\hat{\beta}$ , different SE. `statsmodels:`  
`.get_robustcov_results('HC3')`

**Non-linearity:** True relationship is curved, not a line.

**Fix:** Transform  $X$  (log, polynomial) or use nonlinear models.

**Biased estimators can win:** OLS is best among *unbiased*, but ridge/lasso (from L6) trade a little bias for much less variance.

**Fix:** Regularization when  $p$  is large or predictors are correlated.

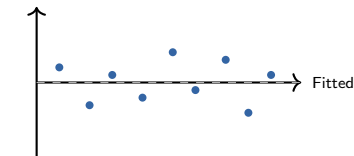
Gauss–Markov says OLS is the best *unbiased linear* estimator.  
It does **not** say OLS is the best estimator, period. (Recall L3:  
bias-variance tradeoff.)

# Part V: Diagnostics

How to check if your model's assumptions hold

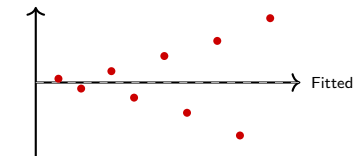
# The Four Diagnostic Plots

Residual **Residuals vs Fitted**



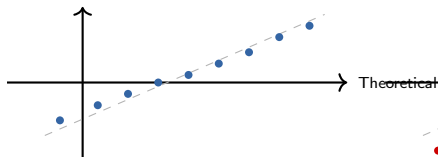
✓ Random scatter = good

**Heteroscedasticity**



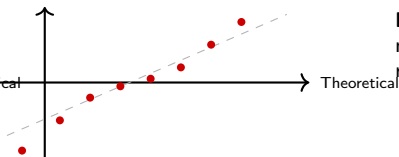
✗ Fan shape = unequal variance

Sample **QQ Plot (Normal)**



✓ Points on line = normal

Sample **QQ Plot (Heavy Tails)**

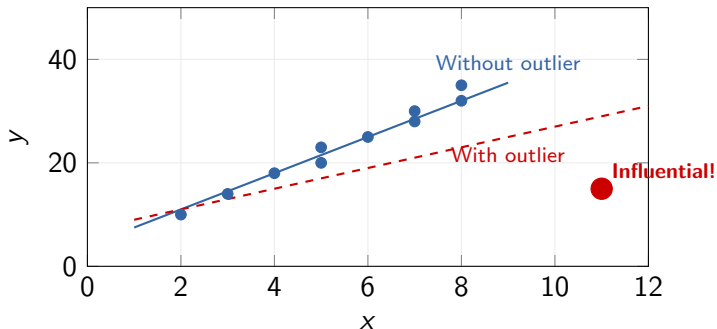


✗ S-shape = non-normal tails

**Top:** Check linearity & homoscedasticity

**Bottom:** Check normality of residuals

## Leverage and Influential Points



**Leverage ( $h_{ii}$ ):** How far  $X_i$  is from  $\bar{X}$ . High leverage = *potential* to influence the fit.  
Rule:  $h_{ii} > 2(p+1)/n$  is high.

**Cook's distance ( $D_i$ ):** How much the fit changes if you remove point  $i$ .  
Rule:  $D_i > 1$  is influential.

# Part VI: Logistic Regression

When  $Y$  is binary: from MLE to inference

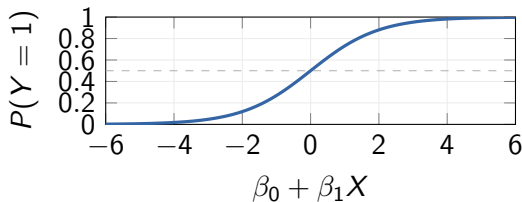
# Logistic Regression as MLE

Connects to L5 (MLE) and L6 (cross-entropy)

$$Y_i \in \{0, 1\}. \quad P(Y_i = 1 \mid X_i) = \sigma(\beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip})$$

$$\text{where } \sigma(z) = \frac{1}{1 + e^{-z}} \quad (\text{logistic / sigmoid function})$$

**No closed-form solution.** Fit by *maximizing the likelihood* (numerical optimization).



The negative log-likelihood is the **cross-entropy loss** from L5. MLE = minimize cross-entropy.



## Inference for Logistic Regression

No exact  $t$ -distribution. Instead, use asymptotic normality of MLE (from L5):

$$Z = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim N(0, 1)$$

SE from the Fisher information matrix:  $\text{SE} = \sqrt{[\mathcal{I}(\hat{\beta})^{-1}]_{jj}}$

**Example:** Predicting college admission ( $Y$ ) from GPA ( $X_1$ ) and test score ( $X_2$ ).

|             | Estimate | SE    | $Z$   | $p$ -value |
|-------------|----------|-------|-------|------------|
| (Intercept) | -12.3    | 2.1   | -5.86 | < 0.001    |
| GPA         | 3.2      | 0.8   | 4.00  | < 0.001    |
| Test score  | 0.015    | 0.005 | 3.00  | 0.003      |

Both GPA and test score are significant predictors of admission.

## Interpreting Logistic Coefficients: Odds Ratios

$$\text{In logistic regression: } \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \dots$$

A 1-unit increase in  $X_j$  multiplies the **odds** by  $e^{\beta_j}$ .

$$\text{Odds Ratio} = e^{\hat{\beta}_j}$$

### Admission example:

**GPA:**  $e^{3.2} = 24.5$

Each extra GPA point  
multiplies admission odds  
by 24.5.

95% CI:  $[e^{1.6}, e^{4.8}] = [5.0, 121.5]$

**Test score:**  $e^{0.015} = 1.015$

Each extra point on the test  
increases odds by 1.5%.

95% CI:  $[e^{0.005}, e^{0.025}] =$   
 $[1.005, 1.025]$

OR > 1: increases odds. OR < 1: decreases odds. OR = 1: no effect.

## OLS vs Logistic: Side by Side

|                     | OLS (Linear)                           | Logistic                                    |
|---------------------|----------------------------------------|---------------------------------------------|
| <b>Response</b>     | Continuous $Y \in \mathbb{R}$          | Binary $Y \in \{0, 1\}$                     |
| <b>Model</b>        | $E[Y] = X\beta$                        | $\log(p/(1-p)) = X\beta$                    |
| <b>Estimation</b>   | Closed form: $(X^\top X)^{-1}X^\top Y$ | Numerical MLE (no closed form)              |
| <b>Distribution</b> | $t_{n-p-1}$ (exact)                    | $N(0, 1)$ (asymptotic)                      |
| <b>SE source</b>    | $\hat{\sigma}^2(X^\top X)^{-1}$        | Fisher info $\mathcal{I}(\hat{\beta})^{-1}$ |
| <b>Coefficient</b>  | $\hat{\beta}_j$ = change in $E[Y]$     | $e^{\hat{\beta}_j}$ = odds ratio            |
| <b>Overall test</b> | $F$ -test                              | LRT ( $-2 \log \Lambda \sim \chi_p^2$ )     |
| <b>Fit metric</b>   | $R^2$ , adj. $R^2$                     | AIC, deviance, pseudo- $R^2$                |

**Same workflow:** Fit model  $\rightarrow$  check significance of each  $\hat{\beta}_j \rightarrow$   
check overall fit  $\rightarrow$  diagnostics.

The tools change (exact  $t$  vs asymptotic  $Z$ ,  $R^2$  vs deviance), but  
the logic is identical.

## Summary: Regression Inference

**Model:**  $Y = X\beta + \varepsilon$ ,  $\varepsilon \sim N(0, \sigma^2 I)$ . Assumptions: LINE.

**SE:**  $SE(\hat{\beta}_j) = \hat{\sigma} \sqrt{(X^\top X)^{-1}_{jj}}$ . Small SE = precise estimate.

**t-test:**  $T = \hat{\beta}_j / SE(\hat{\beta}_j) \sim t_{n-p-1}$ . Tests if predictor  $j$  matters.

**CI:**  $\hat{\beta}_j \pm t_{\alpha/2} \cdot SE$ . Contains 0  $\Leftrightarrow$  not significant.

**Overall  $F$ :** Tests all predictors at once. ANOVA for regression.

**$R^2$ :** Proportion explained. Use adjusted  $R^2$  to compare models.

**Gauss–Markov:** OLS is BLUE. But biased estimators (ridge) can win.

**Logistic:** MLE, Z-test (asymptotic), odds ratios =  $e^{\hat{\beta}_j}$ .

# Practical: Regression Inference in Python

## 1. **OLS regression:**

- ▶ Load a dataset (e.g., `sklearn.datasets.load_diabetes`)
- ▶ Fit with `statsmodels.api.OLS`. Read the summary table
- ▶ Identify significant predictors, interpret  $R^2$ , check overall  $F$

## 2. **Diagnostics:**

- ▶ Plot residuals vs fitted. QQ plot of residuals
- ▶ Compute leverage and Cook's distance. Identify influential points
- ▶ `statsmodels.stats.outliers_influence.OLSInfluence`

## 3. **Logistic regression:**

- ▶ Binary outcome (e.g., `sklearn.datasets.load_breast_cancer`)
- ▶ Fit with `statsmodels.api.Logit`. Compute odds ratios
- ▶ Compare coefficients and CIs with OLS intuition

## Homework

1. Simulated data: generate  $Y = 3 + 2X_1 + 0 \cdot X_2 + \varepsilon$ ,  $\varepsilon \sim N(0, 4)$ ,  $n = 50$ .
  - (a) Fit OLS with both  $X_1$  and  $X_2$ . Which coefficient is significant?
  - (b) Compute 95% CIs. Does the CI for  $\beta_2$  contain 0?
  - (c) What is  $R^2$ ? What is adjusted  $R^2$ ?
2. Using the diabetes dataset in `sklearn`:
  - (a) Fit a multiple regression. Which predictors are significant at  $\alpha = 0.05$ ?
  - (b) Plot residuals vs fitted. Is there evidence of heteroscedasticity?
  - (c) Find the point with the highest Cook's distance. Does removing it change conclusions?
3. Logistic regression: Use the breast cancer dataset.
  - (a) Fit a logistic model with 3 predictors of your choice.
  - (b) Compute odds ratios and their 95% CIs.
  - (c) Which features increase the odds of malignancy?
4. **Theory:** Explain why adding a useless predictor ( $\beta_j = 0$ ) increases  $R^2$  but can decrease  $R^2_{\text{adj}}$ . What does this tell us about model selection?

# Recommended Resources

## **Video: StatQuest — Linear Regression, Clearly Explained**

Step-by-step: fitting,  $R^2$ ,  $p$ -values, and diagnostics. Also: “Logistic Regression.”

## **Interactive: Seeing Theory — Regression Analysis (Brown)**

[seeing-theory.brown.edu/regression-analysis](http://seeing-theory.brown.edu/regression-analysis) — drag points, see how the line and  $R^2$  change.

## **Reading: Wasserman, “All of Statistics” — Ch. 13**

Linear and logistic regression, inference, diagnostics. Concise and rigorous.

## **Reading: James et al., “ISLR” — Ch. 3–4**

Linear regression (Ch. 3) and logistic regression (Ch. 4). Intuitive, with R/Python labs.

## **Python: statsmodels**

OLS, Logit, `.summary()`, `.conf_int()`, `OLSInfluence`, `get_robustcov_results`.

# Questions?

Next: Lecture 13 — Generalized linear models